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Amharic DBpedia Chapter Key Achievement and Deliverable Report

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1. Project Summary

Amharic Wikipedia lacks structured knowledge representation, which limits its integration into global knowledge graphs like DBpedia. This gap hinders the accessibility, discoverability, and interoperability of Amharic content in multilingual semantic web applications.

This project proposes to enhance the **Amharic chapter of DBpedia** by extracting structured data from Amharic Wikipedia. It involves extending the DBpedia Extraction Framework to support Amharic, creating new mappings for infobox templates, and extracting rich semantic data such as citations, disambiguations, and anchor texts.

The output will be published as **RDF triples** and made accessible via a user-friendly web interface, following **FAIR (Findable, Accessible, Interoperable, Reusable)** data principles.

2. Key Achievements and Deliverables

2.1. Wikipedia Templates and Pages

A crucial starting point for structured knowledge extraction is the availability of well-designed templates in Wikipedia. During the project, several new templates were created for the Amharic Wikipedia, and multiple pages were developed using them.

Category	Count	Remark
New Template Created	23	Total: 23 templates Expected to update the existing templates in the future
Updated Templates	0	
New Pages Created	11	Total: 24 pages
Updated Pages	13	

These templates form the foundation for DBpedia mappings by standardizing how entities such as people, books, and organizations are represented.

2.2. Mapping Coverage Progress

Mapping activities progressed significantly compared to the initial status.

✓ Previous coverage: 17 mapped templates

✓ Current coverage: 97 mapped templates

Key improvements:

- ✓ Expansion from 17 → 97 mapped templates, a more than five-fold increase.
- ✓ Growth in Amharic-native template support, strengthening local knowledge representation.
- ✓ Maintained compatibility with English mappings, ensuring continuity, with further translation work planned.

2.3. Mapping Statistics Generation

Mapping statistics provide a quantitative overview of progress and help identify inconsistencies. They measure templates, properties, and their occurrences in Wikipedia pages.

Old statistics report (baseline):

Metric	Count	Total	Percentage
Mapped Templates	20	63	31.75%
Mapped Template Occurrences	1918	3901	49.17%
Mapped Properties	402	2560	15.70%
Mapped Property Occurrences	13216	27836	47.48%

This analysis highlighted template limitations and mapping gaps.

Technical challenge: During statistics generation, templates were detected but properties were ignored, leading to zero results. Investigation revealed reliance on `infobox_test.ttl`, which was not aligned with current dumps.

Fix implemented: The workflow was migrated to `infobox_properties.ttl`, requiring updates in:

- ✓ `CreateMappingStats.scala`
- ✓ `MappingStatsBuilder.scala`

This change allowed templates and properties to be counted correctly, confirming `infobox_properties.ttl` as the reliable source.

Updated Amharic statistics report:

Metric	Count	Total	Percentage	Notes
Mapped Templates	84	84	100%	Achieved full coverage (previously 31.75%)
Mapped Template Occurrences	3958	3958	100%	All templates aligned and fully mapped
Mapped Properties	2392	3095	77.29%	Significant increase from 15.70%
Mapped Property Occurrences	19799	19968	99.15%	Major improvement due to property fixes and normalization

2.4. Documentation

Comprehensive documentation was developed to support contributors. It covers the full workflow:

- ✓ Creating and updating Wikipedia templates
- ✓ Defining mappings to DBpedia ontology
- ✓ Running extractions
- ✓ Generating statistics
- ✓ Querying the knowledge graph

The guide also includes challenges, solutions, and recommendations, enabling researchers and new contributors to replicate and extend the project.

2.5. Website Development

The Amharic DBpedia chapter website was updated to improve accessibility.

Key improvements:

- ✓ Semi-dynamic design: powered by JSON for easier updates.
- ✓ Language support: Amharic, English, and German, with translations managed via JSON files.
- ✓ Reorganized menu: clearer navigation and usability.
- ✓ Refreshed interface: modernized look and improved user experience.

The site now provides a structured entry point for users, mentors, and contributors.

2.6. Amharic DBpedia Template Mapping Automation

Mapping is repetitive and error-prone when done manually. To address this, an automation workflow was prototyped:

- ✓ Template retrieval from Wikipedia.
- ✓ Attribute extraction from raw Wikitext.
- ✓ Automatic translation using machine translation (NLLB or Gemini).
- ✓ Mapping file generation (mapping_am.xml) aligned with DBpedia ontology.
- ✓ Testing and validation using extraction runs and SPARQL queries.

This workflow improves efficiency and consistency in mapping new templates. So far, the first two steps (template retrieval and attribute extraction) have been successfully completed.

2.7. Knowledge Graph Construction

Using the DBpedia Extraction Framework, RDF triples were extracted from Amharic Wikipedia to build a machine-readable Knowledge Graph. The resulting graph supports semantic search, NLP, and linked data integration. The dataset has been pushed here:

<https://github.com/AmharicDBpedia/AmharicDBpedia/tree/GSOC2025/amDbpediaDump/GSoC2025>

2.8. Contribution to the DBpedia Extraction Framework

Key contributions to the DBpedia codebase include:

- ✓ Improved mappings and statistics: Updates submitted as Pull Requests, expanding coverage and correcting issues.
- ✓ New method for statistics generation: Transition to infobox_properties.ttl, enabling reliable local validation with new dumps.

These contributions enhance tooling for contributors, increase mapping coverage, and improve automation reliability for the Amharic DBpedia project.

3. Blog

Throughout the project, progress and milestones have been documented in weekly blog posts. These blogs detail the technical steps, challenges faced, and solutions implemented.

4. Next Steps

Expand mappings to cover more Amharic Wikipedia templates.

- ✓ Finalize and merge PRs for updated statistics and ignore list.
- ✓ Improve automation with LLM/NLLB translation pipelines.
- ✓ Engage the Amharic Wikipedia community for validation and refinement.
- ✓ Work toward DBpedia Live integration for Amharic.

This project lays the foundation for a fully functional Amharic DBpedia chapter, bridging the knowledge gap for underrepresented languages and enriching the global Semantic Web.